Thanh Le

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Website — LinkedIn — GitHub

Education

2022 - Present **PhD in Mathematics**, *University of Nebraska-Lincoln*, Lincoln, NE
Advisors: Xavier Pérez-Giménez and Vinodchandran Variyam
Expected May 2027

2022 - 2023 **M.S. in Mathematics**, *University of Nebraska-Lincoln*, Lincoln, NE

2018 - 2022 **B.S. in Computer Science and Mathematics**, *Fairfield University*, Fairfield, CT
summa cum laude

Positions Held

2025 - Present **Graduate Research Assistant**, *School of Computing, University of Nebraska-Lincoln*, Lincoln, NE

Summer 2026 **Computer Vision Research Intern, Haight Intern Fellow**, *Ocuvera*, Lincoln, NE

2022 - 2024 **Graduate Teaching Assistant**, *Mathematics Department, University of Nebraska-Lincoln*, Lincoln, NE

2020 - 2021 **Software Engineer Intern**, *VietESoft JSC*, Hanoi, Vietnam

Research Interests

AI safety and alignment

Theory of deep learning: transformer expressiveness and chain-of-thought computation

Probabilistic combinatorics: random graph theory

Applied AI: computer vision

Papers and Manuscripts

Accepted Papers

[1] **Thanh Le**, A. Pavan, and N. V. Vinodchandran. *A Tight Hierarchy for Chain of Thought*. Advances in Neural Information Processing Systems (NeurIPS), 2026. To appear.

Submitted Manuscripts

[2] **Thanh Le**, A. Pavan, Xavier Pérez-Giménez, and N. V. Vinodchandran. *Cliques, Quasi-Cliques, and Bicliques in Random Graphs: Sampling and Algorithms*. Submitted, 2026.

Work in Preparation

[3] Derek DeBlieck, **Thanh Le**, and Xavier Pérez-Giménez. *Hamilton cycles with a periodic pattern in random digraphs with minimum total degree at least 2*. In preparation, 2026.

Additional Research Experience

Summer 2026 **Estimating Bed Pressure from Depth Video**
Ocuvera, Lincoln, NE.
Computer Vision Research Intern, Haight Intern Fellow

- Developed deep learning models in PyTorch to estimate bed pressure from depth images, with the goal of supporting pressure injury prevention.
- Developed an automated bed detection and cropping pipeline by training and testing on about 10,000 hospital-room frames.
- Trained and evaluated convolutional neural networks and vision transformers (on 200,000 bed frames) for pressure estimation, comparing performance using mean squared error and structural similarity.

2021 - 2022 **EEG Memory Analysis Using Machine Learning**
Fairfield University, Fairfield, CT.
Research Assistant; supervisors: Dr. Danushka Bandara and Dr. Jessica Karanian

- Investigated machine learning methods for distinguishing true and false memories using EEG recordings from approximately 20 participants.
- Extracted frequency and complexity features from EEG time series and trained classifiers, including random forests and naive Bayes, using scikit-learn.

Honors, Awards, and Fellowships

2026 **Haight Intern Fellowship**
Department of Mathematics, University of Nebraska-Lincoln.
Funding to support graduate students engaged in summer internships.

2024 **Lloyd Jackson Award**
Department of Mathematics, University of Nebraska-Lincoln.
Awarded to one student every year based on scholarship.

2023 **Outstanding Qualifying Exam Award**
Department of Mathematics, University of Nebraska-Lincoln.

2022 **Award for Outstanding Achievement in Mathematics**
Department of Mathematics, Fairfield University.

2021 **William Lowell Putnam Mathematical Competition**
Ranked 193rd out of 2,975 participants.

2019 **Research and Travel Grant**
Fairfield University.
Supported study abroad in Florence, Italy.

2017 **Third Prize, Vietnam Mathematical Olympiad**

Talks

Jun. 2026 *Finding maximum cliques in random graphs.*
Mathematics Seminar, Fairfield University.

Apr. 2026 *How does ChatGPT work under the hood? Transformer explained*.
Graduate Student Seminar, University of Nebraska-Lincoln.

Mar. 2026 *On the existence of Hamilton cycles with a periodic pattern in random digraphs with degree at least 2*.
Graduate Student Combinatorics Conference 2026, University of Illinois Chicago.

May 2025 *On the existence of Hamilton cycles with a periodic pattern in random digraphs with degree at least 2*.
Discrete Mathematics Seminar, University of Nebraska-Lincoln.

Nov. 2024 *Thresholds in random graphs and the Kahn–Kalai conjecture*.
Discrete Mathematics Seminar, University of Nebraska-Lincoln.

Feb. 2023 *Calculus - How and Why*.
Teaching Table, University of Nebraska-Lincoln.

Apr. 2022 *Using Machine Learning methods to study EEG data*.
Fairfield University Research Symposium.

Conferences/Workshops Attended

Oct. 2026 Probabilistic combinatorics at Rutgers, Rutgers University.

May 2026 Recent Developments in Discrete Probability, NITMB, Chicago.

Mar. 2026 Graduate Student Combinatorics Conference 2026, University of Illinois Chicago.

Jun. 2025 SLMath's summer school "Local Limits of Random Graphs", Paris-Saclay University.

Apr. 2025 Lake Michigan Workshop on Combinatorics and Graph Theory, University of Illinois Chicago.

Aug. 2024 Minicourse: Synergies of probabilistic and extremal combinatorics, VIASM, Vietnam.

Jul. 2024 Annual Meeting, VIASM, Vietnam.

Apr. 2024 Lake Michigan Workshop on Combinatorics and Graph Theory, Western Michigan University.

Oct. 2023 AMS Central Sectional Meeting, Creighton University.

Jul. 2021 Blockchain Mathematics and Computing, VIASM, Vietnam.

Teaching Experience

University of Nebraska-Lincoln

2023 - 2024 **Instructor of Record**
MATH 103: College Algebra and Trigonometry (Fall 2024)
MATH 101: College Algebra (Fall 2023, Spring 2024)

2022 - 2023 **Teaching Assistant**
MATH 107: Calculus II (recitations) (Fall 2022, Spring 2023, Summer 2023)
Led recitation sessions twice a week.

Fairfield University

2021 - 2022 **Teaching Assistant**
Machine Learning (Spring 2022)
Theory of Programming Languages (Fall 2021)

Outreach and Service

2026 **Committee Member**
AI Task Force, Mathematics Department, University of Nebraska-Lincoln.
Examine the implications of AI for mathematical research in the department.

2026 **Manuscript Reviewer**
Conference on Neural Information Processing Systems (NeurIPS 2026).

2026 **Graduate School Panelist**
Fairfield University REU.
Spoke about my experiences in mathematics graduate school.

2025 - 2026 **Seminar Co-organizer**
Discrete Mathematics Seminar, University of Nebraska-Lincoln.

2025 **Mathematics Reading Mentor**
Directed Reading Program (DRP), University of Nebraska-Lincoln.
Guided an undergraduate through readings on graph algorithms.

2024 - 2025 **Graduate School Mentor**
Graduate Student Mentor Program, University of Nebraska-Lincoln.
Mentored a first-year math graduate student to support their transition to graduate school.

2024 **Putnam Instructor**
Putnam Seminar, University of Nebraska-Lincoln.

2023, 2024 **Volunteer, Technical Support**
Nebraska Conference for Undergraduate Women in Mathematics (NCUWM), University of Nebraska–Lincoln.

2023, 2024 **Quiz Bowl Moderator**
Math Day, University of Nebraska–Lincoln.

2020, 2022 **Olympiad Mathematics Instructor**
Nguyen Tat Thanh High School for Gifted Students, Kon Tum, Vietnam.
Taught number theory and combinatorics for a week each year. Discussed mathematics careers, Olympiad preparation, and study abroad opportunities.

2021 - 2022 **Secretary and Treasurer**
Pi Mu Epsilon, Fairfield University Chapter.

2021 **Website Developer**
Hanoi Math Modeling Competition, Hanoi, Vietnam.
Led the development of the competition website from June to October 2021.

2019 **Competition Grader**
Regional MATHCOUNTS competition, Fairfield University.

Technical Skills

Programming: Python, MATLAB, Mathematica, SQL, C#, Java, Haskell, Lean

Machine learning and data analysis: PyTorch, TensorFlow, scikit-learn, pandas, Matplotlib

Methods: computer vision, image segmentation, transfer learning, Monte Carlo simulation

Other tools and frameworks: LaTeX, ASP.NET Core, React

Additional Training

2026 **Quantitative Finance**
The Erdős Institute.

2025 **Neural Networks and Deep Learning**
DeepLearning.AI, via Coursera.

Languages

Vietnamese (native); English (fluent)